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Dynamic Governance Technology for Power Knowledge Warehouse Based on Multi-System Data Collaboration under Microservice Architecture

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***Abstract*—This paper proposes a dynamic governance technology for power knowledge warehouse based on multi-system data collaboration within a microservice architecture. This method constructs a power knowledge warehouse representation model using power entities, business relationships, operational events, temporal states, and knowledge versions as unified objects. Data access services, semantic parsing services, knowledge construction services, conflict detection services, dynamic governance services, and query reasoning services are designed to achieve loosely coupled access and continuous governance of cross-system data. To address the risks of cross-system attribute conflicts, state drift, and version overwrite, a conflict resolution algorithm based on source credibility, time freshness, field stability, and historical hit rate is further proposed, and a consistency optimization objective function for incremental updates is established. Experimental results show that, in scenarios with continuous access to multi-source power data, the proposed method outperforms centralized ETL, rule-based governance, and ordinary graph database methods in terms of data consistency, knowledge update latency, and conflict resolution accuracy. The research results demonstrate that combining microservice architecture with knowledge graph governance mechanisms can effectively improve the dynamic maintenance capabilities and cross-system collaboration efficiency of the power knowledge repository.**

Keywords- Microservice architecture; power knowledge repository; data collaboration; knowledge graph; dynamic governance; consistency optimization

I. INTRODUCTION

The construction of new power systems is driving continuous changes in power grid operation modes, business collaboration models, and data management models. Dispatch automation systems generate power flow sections, equipment status, and alarm events; asset management systems maintain equipment ledgers, commissioning records, and lifespan information; operation and maintenance systems accumulate defect work orders, maintenance measures, and on-site handling processes; marketing systems store user-side loads, service requests, and transformer area relationships; metering systems provide data collection terminals, meter status, and abnormal metering information; and new energy systems continuously generate inverter, energy storage equipment, distributed power sources, and grid connection status data. In recent years, knowledge graphs have been used for tasks such as power fault diagnosis, handling recommendations, and data quality verification, capable of expressing complex relationships between power objects in the form of entities, relationships, and events [1-3]. Simultaneously, knowledge graphs are also used in higher-level data governance frameworks, emphasizing data quality, semantic constraints, traceability, and interpretability [4]. Existing research on microservice data management indicates that cross-database transactions, consistency maintenance, and asynchronous compensation mechanisms in distributed services are key factors affecting system reliability and throughput [5-7]. This paper focuses on solving three key problems: first, how to map multi-system power data into unified knowledge units; second, how to automatically detect, score, and resolve cross-system conflicts; and third, how to incrementally evolve knowledge versions while maintaining consistency in a microservice environment.

II. POWER MULTI-SYSTEM DATA COLLABORATION REQUIREMENTS AND PROBLEM MODELING

A. Multi-Source Power Data Collaboration Objects

The collaboration objects addressed in this paper include equipment entities, business relationships, operational events, state sequences, and version records. The goal of multi-system collaboration is not simply to merge fields, but to enable data describing the same object, state, or event in different systems to be mapped to a unified knowledge space, and to maintain traceability, consistency, and queryability during continuous changes [8].

Let the set of power business systems be:

$$\mathcal{S} = \{S_1, S_2, \dots, S_m\} \tag{1}$$

Where S_i represents the i business system, and m represents the number of systems participating in the collaboration. Each system contains a local entity set, a field set, an event set, and a state set.

B. Unified Representation of the Power Knowledge Repository

This paper defines the power knowledge repository as a six-tuple containing entities, relations, attributes, time, version, and mapping functions:

$$\mathcal{K} = (E, R, A, T, V, \phi) \tag{2}$$

Where E represents the set of power entities, R represents the set of relationships between entities, A represents the set of

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entity attributes, T represents the time dimension, V represents the set of knowledge versions, and ϕ represents the cross-system semantic mapping function.

The unified knowledge warehouse is formally defined as:

$$KW = \langle E, R, A, O, T, V, \Phi, \Omega \rangle \tag{3}$$

where E denotes the set of power entities, R denotes business relationships, A denotes attributes, O denotes operational events, T denotes temporal states, V denotes knowledge versions, Φ denotes the cross-system semantic mapping function, and Ω denotes ontology and schema constraints.

The five unified object types are defined as follows:

$$\begin{aligned} e &= \langle id, \text{type}, A_e \rangle \\ r &= \langle e_i, e_j, \text{type}, t \rangle \\ o &= \langle e, \text{type}, \text{content}, t, s \rangle \\ q &= \langle e, a, \text{value}, t \rangle \\ v &= \langle id, \Delta K, t, \text{operator} \rangle \end{aligned} \tag{4}$$

where e is a power entity, r is a business relationship, o is an operational event, q is a temporal state, and v is a knowledge version. Schema evolution is handled by recording new field mappings and ontology changes in Ω.

For a data object x_i from system S_i, its unified knowledge mapping process is as follows:

$$e = \phi(x_i, S_i) \tag{5}$$

Here, e represents the mapped unified knowledge entity. If objects from multiple systems are mapped to the same entity, the system continues to determine whether their attribute values, state values, and event descriptions are consistent.

C. Data Conflicts and Consistency Constraints

Define the attribute values of entity e in systems S_i and S_j as $a_i(e)$ and $a_j(e)$, respectively. The conflict detection function is:

$$C(e) = \begin{cases} 1, & a_i(e) \neq a_j(e) \\ 0, & a_i(e) = a_j(e) \end{cases} \tag{6}$$

When $C(e) = 1$, it indicates that the entity has data conflicts in at least two systems, requiring a dynamic governance process. Further, the knowledge warehouse consistency score is defined as:

$$G = 1 - \frac{N_c}{N_a} \tag{7}$$

Where N_c represents the number of conflicting knowledge units, and N_a represents the total number of knowledge units participating in the collaboration. The closer G is to 1, the higher the consistency of the knowledge warehouse.

III. Dynamic Governance Framework for Power Knowledge Warehouse under Microservice Architecture

A. Overall Architecture Design

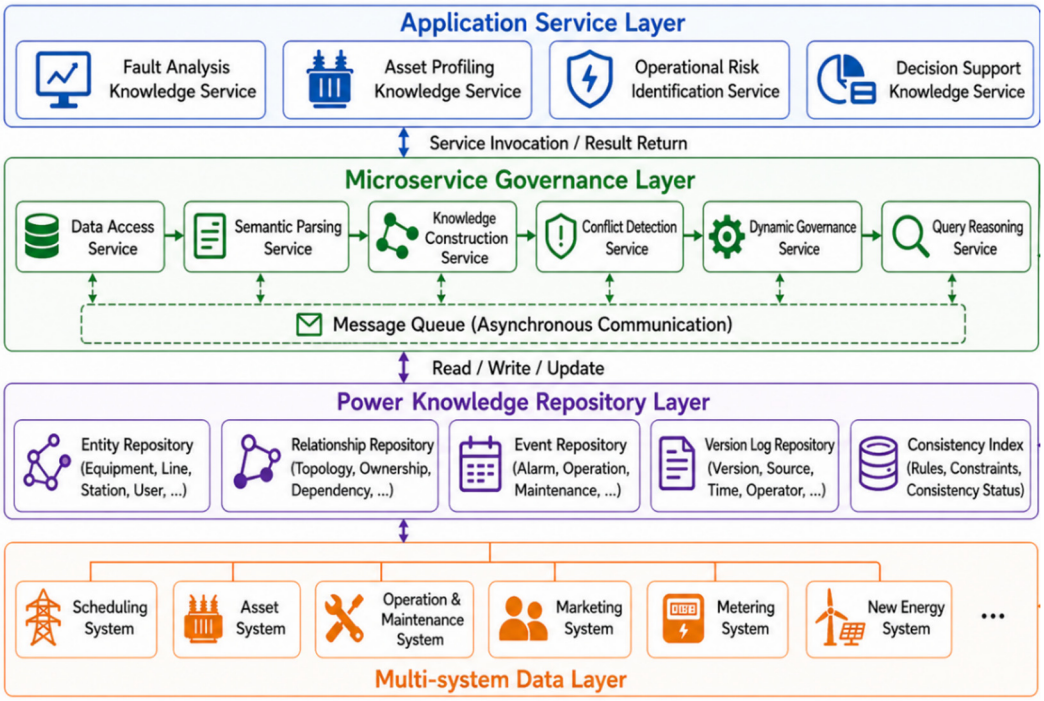


Figure 1. Dynamic Governance Framework of Power Knowledge Repository under Microservice Architecture.

The dynamic governance framework proposed in this paper is shown in Figure 1. The framework includes a multi-system data layer, a microservice governance layer, a power knowledge repository layer, and an application service layer [9]. The multi-system data layer is responsible for accessing data from scheduling, asset, operation and maintenance, marketing,

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metering, and new energy systems; the microservice governance layer consists of data access services, semantic parsing services, knowledge construction services, conflict detection services, dynamic governance services, and query reasoning services; the knowledge repository layer stores entities, relationships, events, version logs, and consistency indexes; and the application service layer provides knowledge services for fault analysis, asset profiling, operational risk identification, and decision support [5,6].

B. Data Access and Event Encapsulation Mechanism

The data access service is responsible for identifying the data format, field type, object identifier, and timestamp of different systems. To reduce service coupling, the access results are encapsulated as knowledge events:

$$q = (id, type, value, t, source) \tag{8}$$

Wherein, id represents the object identifier, $type$ represents the event type, $value$ represents the event content, t represents the timestamp, and $source$ represents the data source system.

C. Semantic Parsing and Knowledge Unit Generation

The semantic parsing service maps fields, codes, and text descriptions from different systems to a unified knowledge schema. This paper uses name similarity, attribute similarity, and relational neighborhood similarity to jointly determine the entity mapping results:

$$Sim(e_i, e_j) = \beta_1 Sim_n + \beta_2 Sim_a + \beta_3 Sim_r \tag{9}$$

Where Sim_n represents name similarity, Sim_a represents attribute similarity, Sim_r represents relation neighborhood similarity, and β_1, β_2 and β_3 are weighting coefficients.

D. Dynamic Governance and Version Evolution Mechanism

After receiving conflict detection results, the dynamic governance service performs a comprehensive score based on data source credibility, update time, field stability, and historical hit rate. If the new data has high credibility, the main version of the knowledge repository is updated; if the credibility is insufficient, it enters the candidate version area for further verification. The knowledge version evolution process is represented as follows:

$$V_{t+1} = \alpha V_t + (1 - \alpha)\Delta V_t \tag{10}$$

Where V_t represents the current knowledge version, ΔV_t represents the incremental update content in this round, and α represents the historical version retention coefficient.

IV. Multi-system Data Collaboration Consistency Optimization Method

A. Conflict Resolution Algorithm Based on Credibility Weights

In multi-system data collaboration, simply adopting "latest data priority" or "master system priority" can easily lead to erroneous coverage [10]. For each candidate knowledge unit k_i, the conflict resolution score is calculated using four normalized factors: source credibility $C_s(k_i)$, time freshness $F_t(k_i)$, field stability $S_f(k_i)$, and historical hit rate $H_h(k_i)$. The aggregate score is defined as:

$$\begin{gathered}\text{Score}(k_i) = w_1 C_s(k_i) + w_2 F_t(k_i) + w_3 S_f(k_i) + w_4 H_h(k_i) \\ w_1 + w_2 + w_3 + w_4 = 1\end{gathered} \tag{11}$$

The time freshness factor is calculated as:

$$F_t(k_i) = \exp\left[-\lambda(t_{\text{now}} - t_i)\right] \tag{12}$$

where t_i is the update time of candidate unit k_i, t_{now} is the current governance time, and λ is the time decay coefficient. $C_s(k_i)$ is assigned according to the authority level of the source system, $S_f(k_i)$ is determined by the historical volatility of the corresponding field, and $H_h(k_i)$ is calculated from the proportion of previous governance decisions verified as correct. In this study, the weights are manually tuned on a validation subset as $w_1 = 0.35, w_2 = 0.25, w_3 = 0.20$, and $w_4 = 0.20$, reflecting the higher importance of source authority in power enterprise data governance.

B. Consistency Optimization Objective Function

Multi-system data collaboration not only requires a low conflict rate but also controllable update latency and acceptable governance overhead. This paper defines the comprehensive optimization objective function as:

$$\min \mathcal{L} = \lambda_1 C_r + \lambda_2 D_t + \lambda_3 U_c \tag{13}$$

Where C_r represents the knowledge warehouse conflict rate, D_t represents the data update latency, U_c represents the knowledge update cost, and λ_1, λ_2 and λ_3 are weighting coefficients. This objective function can comprehensively evaluate governance quality, real-time performance, and system overhead. In actual operation, the weight of λ_2 can be increased for high-time-efficiency scenarios such as scheduling alarms and fault handling; the weight of λ_1 can be increased for high-accuracy scenarios such as asset ledgers and equipment history.

V. Experimental Design and Result Analysis

A. Experimental Environment and Dataset Construction

The experiment constructed a semi-simulation dataset containing six types of power business systems: 1.2 million records of dispatch operation data, 350,000 records of equipment asset data, 180,000 records of operation and maintenance work orders, 800,000 records of marketing user data, 1.5 million records of metering and data collection data, and 450,000 records of new energy operation data [11]. The knowledge warehouse contained 216,000 entities, 843,000 relationships, and 3.1 million state events. The experimental server was configured with a 16-core CPU, 64 GB of memory, and 2 TB of SSD, with 12 microservice containers. Comparison methods included centralized ETL, rule-based governance, ordinary graph database methods, and the proposed DG-KW

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method. Evaluation metrics included data consistency score, average update latency, and conflict resolution accuracy.

B. Data Consistency Improvement Effect

Figure 2 shows the consistency changes of different methods during continuous data access. It can be seen that the centralized ETL method can maintain basic consistency in the early stages, but its consistency improvement gradually slows down as incremental data increases; the rule-based governance method is constrained by fixed rules and fluctuates significantly in complex conflict scenarios; the ordinary graph database method has good relational expression capabilities but lacks a dynamic conflict resolution mechanism; the method proposed in this paper tends to stabilize after the 10th batch, with a final consistency of 0.962. This result indicates that semantic mapping, credibility scoring, and version candidate mechanisms can jointly reduce the cross-system conflict rate.

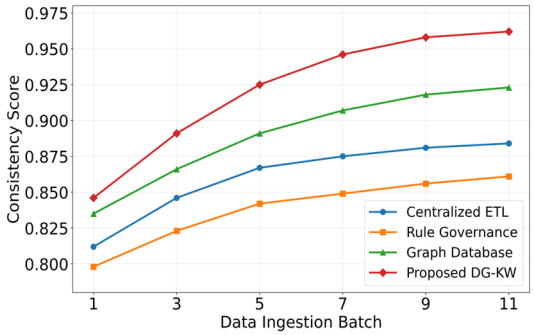


Figure 2. Trends in Data Consistency of the Power Knowledge Warehouse under Different Methods.

C. Analysis of Knowledge Incremental Update Latency

Table 1 shows the average update latency of different methods as the scale of incremental data increases. Experimental results show that the latency of the centralized ETL method increases rapidly with the increase of data scale, reaching an average latency of 8.7 seconds with 1 million incremental data entries; the rule governance method has slightly lower latency, but the rule matching overhead is large in conflict scenarios; the ordinary graph database method has certain advantages in relation updates, but lacks service-level task decomposition; the method in this paper controls the latency to within 3.1 seconds through microservice decoupling, asynchronous message queue processing, and incremental version update mechanism. This result verifies the role of microservice architecture in improving the efficiency of dynamic knowledge governance.

TABLE I. COMPARISON OF KNOWLEDGE INCREMENTAL UPDATE LATENCY

Incremental data size (in ten thousand records)	Centralized ETL	Rule-based governance	Graph Database	This article's method
10	1.8	1.5	1.3	0.9
30	3.2	2.8	2.4	1.5
50	5.1	4.2	3.5	2.1
70	6.9	5.6	4.4	2.6
100	8.7	7.3	5.8	3.1

VI. CONCLUSION

This paper addresses the dynamic governance problem of power knowledge warehouses in a microservice architecture, focusing on multi-system data collaboration. It proposes a technical approach integrating data access, semantic mapping, knowledge construction, conflict detection, version evolution, and consistency optimization. This method improves the efficiency of multi-system data access through microservice decoupling, enhances the power knowledge representation capability through entity-relationship-event-version joint modeling, and improves the accuracy of cross-system data governance through a credibility-weighted conflict resolution algorithm. Experimental results show that the proposed method outperforms traditional methods in terms of data consistency, knowledge update latency, and conflict resolution accuracy, and can adapt to the application requirements of continuous growth and dynamic changes in power multi-system data.

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